Web Application Frameworks
CS457
Fall 2025

Assignment # 2

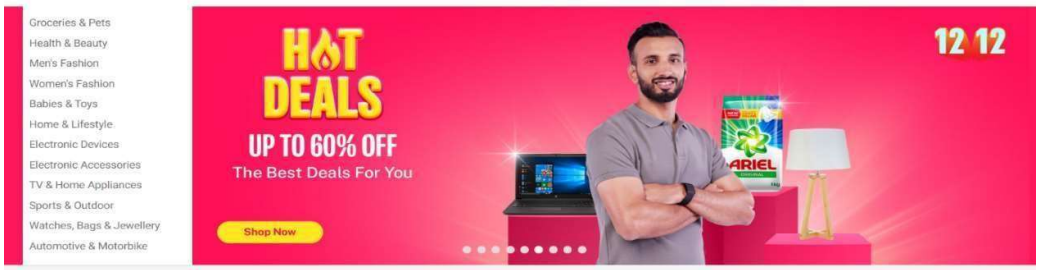
Deadline: 31st December 2025 before 15:00

Note: This is a group assignment. A group can comprise a maximum of two members. Name your zip file as WAF-A2-YourNames.zip/rar. Submit your assignment in the fs folder "WAF-F25-A2". Cheating will be penalized. There is zero tolerance for cheating. Please follow the deadline. Submit it before the deadline. The deadline will not be extended. Please start from day 1.

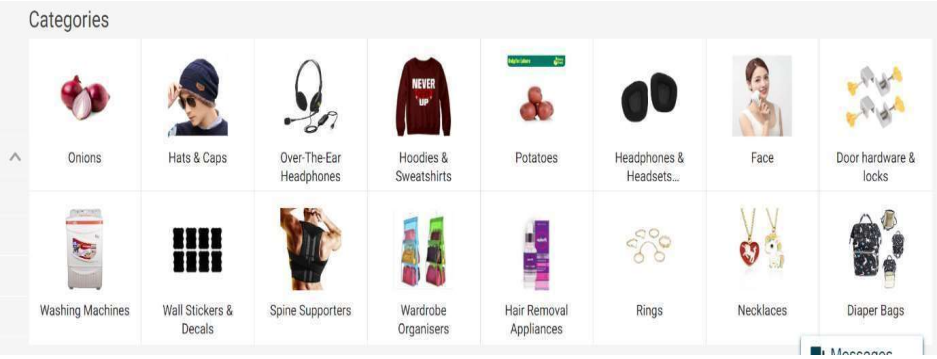
You must create a replica of Daraz.com. Create a complete e-commerce website using ReactJS, MongoDB, Mongoose, ExpressJS, and NodeJS. Make sure to exhaust all CRUD operations when interacting with the database. The e-commerce site should have the following features:

1. Home page: The home page should include the following features:
   - **Navigation Bar:** The navigation bar should include a login, signup, about, and contact us page.
   - **Search and View Cart Facility:** The home page should include a search text field, a button to search among various products of different categories, and a view cart button.

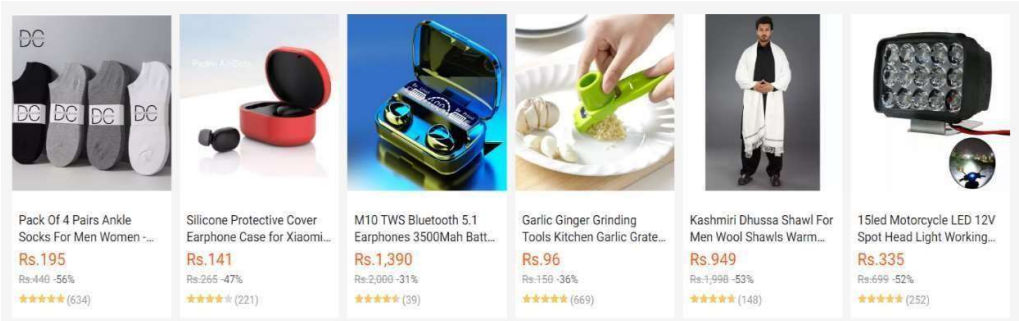
   - **Side Panel and Ad ticker:** The left side panel should include ten clickable categories of products, and the right side should include at least ten clickable advertisements of ten companies, such as follows. The dots on the ad panel show one advertisement, and upon clicking the dots, the advertisement should change accordingly.



   - **Bottom part of the home page:** This part contains two sections shown as follows:



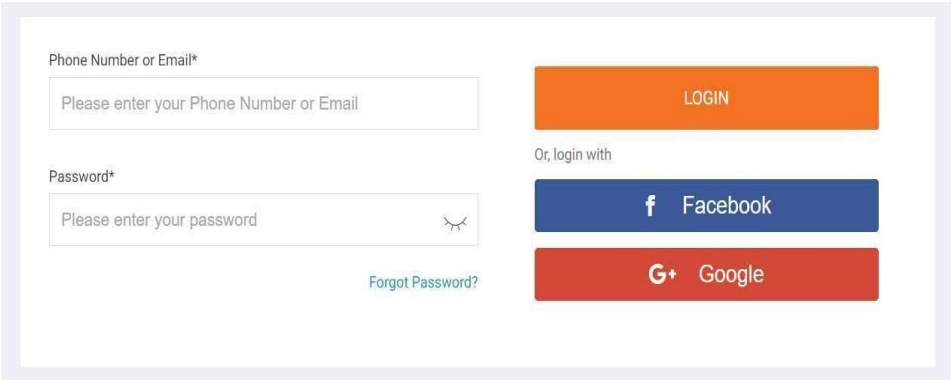
The upward arrow should scroll up to the advertisement and navigation bar section. Random products of different categories should be selected. The image of the product and the title of the product should be read from MongoDB.



There should be an upward arrow as shown previously, and this should scroll up to the advertisement and navigation bar section. Random products of different categories should be selected. The image, title, price, and manual rating of the product should be read from MongoDB. If you are getting from the API, you should store it in MongoDB and then access it from there.

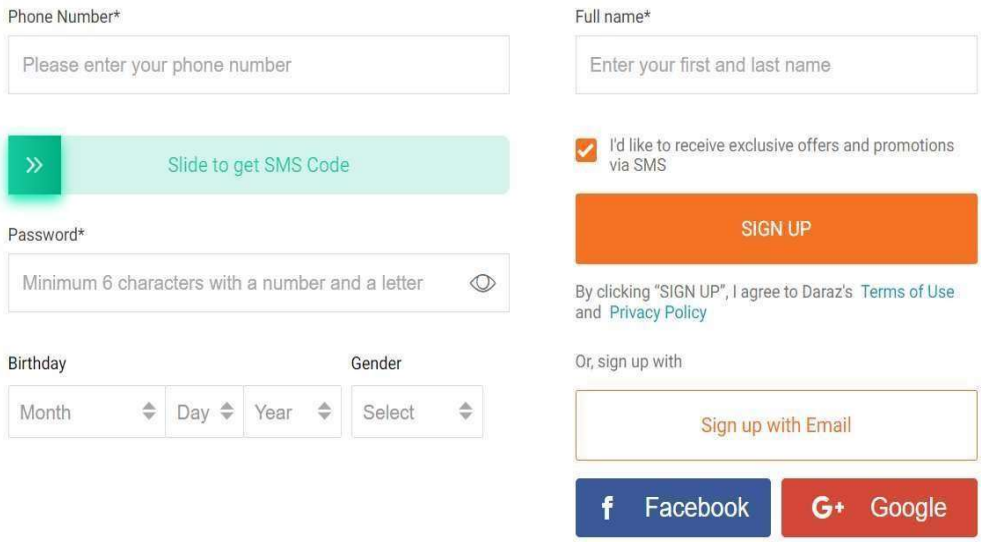
2. Login Page: Login can be accessed through Facebook, Gmail, or an account. All entries should be recorded in the database to utilize it effectively.

Welcome to E-Mart



Perform all possible validation checks. On clicking Facebook, a separate window asking for Facebook credentials should pop up. On clicking Google, a separate window asking for Gmail credentials should pop up. For reference, you can visit the Daraz login page and get an idea.

3. **Signup Page:** The signup page should include the following, and ensure interaction is made with the database.



4. **Product detail:**

5. **Shopping cart and User profiles/history:** Similarly, for shopping carts, user profiles, and past data of users, you can visit the Daraz site to get an idea.

Try to treat this assignment as a product. You should think about completing this project in all aspects to give yourself a chance to exercise maximum practice to get hands-on experience.